

Olive the World!

Computer Science Senior Capstone

Zachary Ciappa, Elysee Lucia Jacinto, Sabrina Shi

Project Plan:

Our project will consist of a story game that follows an olive's journey around the world. We noticed a lack of new, engaging games that expose children to diverse cultures. Our game aims to showcase different cultures to players in an exciting way: food. Something that all three of us enjoy is eating, which is a major way that culture is spread and appreciated. The story starts with an olive, fresh off the tree, that wants to become the best chef in the world. In order to become the best chef, you will need to master other cultures' culinary traditions.

We plan to start this project to reflect three countries: Mexico, Italy, and the Philippines. The Mexico game is a whack-a-mole inspired game, where you must hit the pinatas that contain candy, but avoid the ones that contain the poisonous spicy chiles. The Italy game is a pipe-mania inspired game, where you must fix the flow of the pasta sauce to the corresponding plate by changing the orientation of the rigatoni pipes. The Philippines game is a *Crossy Road* inspired game, where you must traverse the streets of Manila, collecting lumpia ingredients while avoiding traffic and various hazards.

The main programming language used will be JavaScript to control all the gameplay logic. HTML will be used to create the game window and load the JavaScript files. Visual Studio Code (VS Code) will be used to write and organize the code, GitHub will be used to save versions of the project and collaborate with each other, GitHub Pages will be used for web hosting so anyone can play the game, and Piskel will be used for the art/sprite design. Phaser, a JavaScript library, will be used to manage physics, input, animations, and scene transitions, making it easier to build and run the game.

Olive the World! will be a groundbreaking game that aims to foster diversity in children's games through fascinating discovery of the world and its food. In addition to the three games previously mentioned, we hope to develop three more mini games on the cuisine of China, Brazil, and the United States. All of the materials we will be using for this project is either open source or made by us. The work will be split into three, with each team member taking on one game. Additionally, Ella will be responsible for art, Zac will be responsible for the opening/closing sequences, and Sabrina will be responsible for the inter-game sequences.

Languages/Software Needed:

Languages:

- JavaScript
- HTML

Software:

- Visual Studio Code
- GitHub/GitHub Pages
- Piskel